

17/2/Jul/24 2:57:44 PM \\saalibkgpds01a\Projects\Projects\WPEC\Projects\303011_Infra\31011-02282\2_Dwg\12.11.05_Working\WAS\CWA_COMMON\Civil\CONCRETE STANDARD\MD-520-1000-EG-CV-STD-0001.C01.dgn

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1.0 GENERAL

- ALL WORKS SHALL BE UNDERTAKEN IN ACCORDANCE WITH THE LATEST EMPLOYER APPROVED, SIGNED AND STAMPED EDITIONS OF RELEVANT STANDARDS AND THEIR AMENDMENTS, PROJECT SPECIFICATIONS AND THE DESIGN AND STANDARD DRAWINGS AS SPECIFIED.
- ALL DIMENSIONS ARE GIVEN IN MILLIMETERS UNLESS NOTED OTHERWISE.
- DESIGN DRAWINGS SHALL NOT BE SCALED AND ALL DIMENSIONS SHALL BE READ DIRECTLY FROM DESIGN DRAWINGS. ANY DISCREPANCIES WITHIN THE DOCUMENTATION SHALL BE REFERRED TO THE EMPLOYER FOR CLARIFICATION BEFORE PROCEEDING.
- PRODUCTS EQUIVALENT TO THOSE SPECIFIED ON THE DESIGN DRAWINGS MAY BE USED ONLY WITH APPROVAL OF THE EMPLOYER.
- FOR PROJECT SPECIFIC NOTES REFER TO DESIGN DRAWINGS.
- ALL LEVELS ARE GIVEN IN METRES UNLESS NOTED OTHERWISE.
- VERTICAL DATUM.
RAK AREA.
PLANT BENCHMARK ELEVATION - 100.00 EQUIVALENT TO ELEVATION (MSL) - 7.0 (COMPANY REFERENCE DATUM TO MSL)
WAS AREA.
PLANT BENCHMARK ELEVATION - 100.00 EQUIVALENT TO ELEVATION (MSL) - 846.00 (COMPANY REFERENCE DATUM TO MSL)
- HORIZONTAL GRID PROVIDED BASED ON UTM CO-ORDINATE SYSTEM.
- ALL FOUNDATION EXCAVATIONS SHALL BE INSPECTED AND APPROVED BY THE SITE ENGINEER PRIOR TO PLACING OF CONCRETE.
- ALL STRUCTURAL DRAWINGS SHALL BE READ IN CONJUNCTION WITH ALL RELEVANT ARCHITECTURAL, ELECTRO-MECH'L & PLUMBING DRAWINGS.

2.0 EXCAVATION AND FOUNDATIONS

- EXCAVATION AND FOUNDATION PREPARATION SHALL COMPLY WITH STANDARD SPECIFICATION MD-520-1000-EG-CV-SPC-0001.
- NO BACKFILLING SHALL COMMENCE AGAINST WALLS UNTIL 28 DAYS HAVE ELAPSED FROM COMPLETION OF THE CONCRETE POUR OR APPROVED PROPPING HAS BEEN INSTALLED.
- ANY OVER EXCAVATION AT FOOTING LOCATIONS SHALL BE MADE GOOD WITH SELECT FILL COMPACTED TO AT LEAST THE BEARING CAPACITY OF THE SURROUNDING UNDISTURBED SOIL IN ACCORDANCE WITH STANDARD SPECIFICATION MD-520-1000-EG-CV-SPC-0001. ALTERNATIVELY LEAN MIX CONCRETE MAY BE USED.
- SOIL UNDER SLABS ON GRADE SHALL BE COMPACTED TO A MINIMUM DEPTH OF 300 mm TO ACHIEVE 95% MODIFIED MAXIMUM DRY DENSITY AS MEASURED BY ASTM D1557 FOR SANDS AND LABORATORY TEST ASTM D2922 FOR COHESIVE SOIL.
- SOIL UNDER FLAT BOTTOMED TANKS SHALL BE COMPACTED AS PER NOTE 2.4 TO ACHIEVE 98% MODIFIED MAXIMUM DRY DENSITY.
- ALL EXCAVATIONS FOR FOOTINGS SHALL HAVE A MINIMUM THICKNESS OF 75mm OF LEAN MIX CONCRETE (GRADE IN ACCORDANCE WITH STANDARD SPECIFICATION) APPLIED IMMEDIATELY ON COMPLETION OF THE EXCAVATION. NO EXCAVATION SHALL BE LEFT OPEN OVERNIGHT WITHOUT POURING LEAN MIX CONCRETE.
- TO FACILITATE DRAINAGE AWAY FROM CONCRETE STRUCTURES THE CONCRETE CONTRACTOR SHALL BACKFILL AND COMPACT LOCALLY BEHIND ALL KERBS, AND AROUND THE EDGES OF ALL SLABS AND ISOLATED PEDESTALS, TO THE DETAILS SHOWN ON THE DESIGN DRAWINGS AND AS DIRECTED BY THE EMPLOYER.
- NO BACKFILL SHALL BE PLACED AGAINST NEW CONCRETE FOR SEVEN DAYS AFTER IT IS POURED FOR SERVICE TUNNELS AND TRENCHES.
- CONCRETE SURFACES BELOW GROUND SHALL BE COATED WITH TWO (2) COATS OF BITUMINOUS PAINT.

3.0 FORMWORK

- ALL FORMWORK AND FALSEWORK SHALL BE IN ACCORDANCE WITH STANDARD SPECIFICATION MD-520-1000-EG-CV-SPC-0002 AND SHALL COMPLY WITH ACI 347.
- ALL EXPOSED EDGES OF CONCRETE SHALL BE PROVIDED WITH A 20mm x 20mm CHAMFER OR FILLET.
- FORM REMOVAL SHALL BE IN ACCORDANCE WITH ACI 301 AND STANDARD SPECIFICATION MD-520-1000-EG-CV-SPC-0002.

4.0 CONCRETE

- ALL CONCRETE WORKS SHALL BE IN ACCORDANCE WITH STANDARD SPECIFICATION MD-520-1000-EG-CV-SPC-0002 AND ACI 301.
- CONCRETE FINISHES SHALL COMPLY WITH ACI 302.1R, STANDARD SPECIFICATION MD-520-1000-EG-CV-SPC-0002 AND AS FOLLOWS, UNLESS NOTED OTHERWISE ON THE DESIGN DRAWINGS:
- CONCRETE FINISHES AND TOLERANCES SHALL BE IN ACCORDANCE WITH STANDARD SPECIFICATION MD-520-1000-EG-CV-SPC-0002, AND SHALL COMPLY WITH ACI 301 & ACI 117.
- CONCRETE CURING SHALL COMPLY WITH STANDARD SPECIFICATION MD-520-1000-EG-CV-SPC-0002.
- FOR CONCRETE STRENGTH GRADES REFER STANDARD SPECIFICATION MD-520-1000-EG-CV-SPC-0002. DESIGNER SHALL SPECIFY CONCRETE GRADE ON DESIGN DRAWING.

THE CONCRETE STANDARD NOTES SHOWN ON THIS DRAWING APPLY UNLESS NOTED OTHERWISE ON THESE STANDARD DRAWINGS OR ON THE DESIGN DRAWINGS

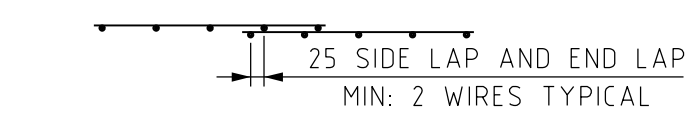
ALL CONCRETE STRUCTURES INCLUDING CONCRETE WITH MIN.REINFORCEMENT FOR TEMPERATURE AND SHRINKAGE CONTROL	C35	42db	63db	23
ALL CONCRETE ON MARINE STRUCTURES, AND OUTFALL STRUCTURES	C40	39db	59db	20
CONCRETE FOR SLABS ON THE GRADE/ SUMP PITS	C35	42db	63db	18
REINFORCED CONCRETE FOR WATERFRONT STRUCTURES	C35	42db	63db	18
REINFORCED STRUCTURES OVER SEAWATER WHICH ARE 5 METERS ABOVE MEAN HIGH WATER SPRINGS	C30	45db	68db	18
DRAINAGE AND UTILITY STRUCTURES, DUCT BANKS	C25	50db	74db	18
ALL PRE-STRESSED CONCRETE	C35	42db	63db	18
ALL PRE-CAST CONCRETE	C35	42db	63db	18
ALL MASS (UNREINFORCED) CONCRETE INCLUDING BLINDING,DUCT BANKS ETC. NOT EXPOSED TO ATMOSPHERIC CONDITIONS.	C15	-	-	-

* db = BAR DIAMETER * COMP L_d SHALL BE REDUCED BY A FACTOR OF (A_g Req/A_{st} PROV)

- ADDITIVES SHALL NOT BE USED WITHOUT PRIOR APPROVAL OF THE EMPLOYER.
- CONSTRUCTION JOINTS AND SLEEVES SHALL NOT BE USED IN POSITIONS OTHER THAN THOSE INDICATED ON THE DESIGN DRAWINGS WITHOUT PRIOR APPROVAL OF THE EMPLOYER.
- A VAPOUR BARRIER MEMBRANE CONSISTING OF A 0.25mm POLYTHENE FILM SHALL BE PLACED UNDER ALL SLABS POURED ON GROUND CONFORMING TO ASTM D2103. LAP VAPOUR BARRIER MEMBRANE MINIMUM 100mm AND SEAL WITH ADHESIVE TAPE TO ENSURE A COMPLETE MOISTURE BARRIER.
- THE MAXIMUM WATER-CONTENTITIOUS MATERIALS RATIO SHALL BE 0.4 FOR LIQUID-RETAINING (HYDRAULIC) STRUCTURES AND 0.5 FOR ALL OTHER CONCRETE, UNLESS OTHERWISE SPECIFIED IN THE CONTRACT DOCUMENTS.
- THE MAXIMUM FREE FALL OF CONCRETE DURING PLACEMENT SHALL NOT EXCEED 2 METERS.
- PORTLAND CEMENT SHALL CONFORM TO ASTM C150, TYPE 1 OR TYPE II, UNLESS OTHERWISE SPECIFIED ON THE DESIGN DRAWINGS. ONLY ONE BRAND OF CEMENT SHALL BE USED. FLY ASH MAY BE USED IN ACCORDANCE WITH ACI 301 AND SHALL CONFORM TO ASTM C618, CLASS F, UNLESS OTHERWISE SPECIFIED ON THE DESIGN DRAWINGS.

5.0 REINFORCEMENT

- FOR MINIMUM CHARACTERISTIC STRENGTH FOR REINFORCEMENT AS PER PROJECT SPECIFICATIONS.
- REINFORCING STEEL BAR SHALL BE DEFORMED BAR CONFORMING TO MD-520-1000-EG-CV-SPC-0002 AND ASTM A615/A615M, GRADE 60 AND MINIMUM SPECIFIED CHARACTERISTIC STRENGTH F_y = 420 N/MM² UNLESS NOTED OTHERWISE.
- WELDED STEEL FABRIC SHALL CONFORM TO A185/A185M OR ASTM A497.
ALL FABRIC SHALL BE SUPPLIED IN FLAT SHEETS. LAP FABRIC AS SHOWN BELOW.



- LOCATION OF SOME SPLICES ARE SHOWN ON DRAWINGS. SPLICES NOT SHOWN SHALL BE IN ACCORDANCE WITH THE REQUIREMENTS OF CONCRETE REINFORCING STEEL INSTITUTE (CRSI) MANUAL OF STANDARD PRACTICE AND AS APPROVED BY MAADEN REPRESENTATIVE.
- ALL SPLICING OF REINFORCING BARS SHALL BE DONE BY LAPPING UNLESS OTHERWISE APPROVED. BARS SPLICED BY NON CONTACT LAP SPLICES IN FLEXURAL MEMBERS SHALL NOT BE SPACED TRANSVERSELY FURTHER APART THAN ONE-FIFTH THE REQUIRED SPLICE LENGTH NOR 150mm.
- ALL TENSION LAP SPLICES SHALL BE CLASS B AS DEFINED IN ACI 318.TENSION LAP SPLICE LENGTH SHALL BE 13LD WHERE LD IS THE TENSILE DEVELOPMENT LENGTH IN ACCORDANCE WITH ACI WITH THE MODIFICATION FACTOR FOR EPOXY COATING OR REFER ABOVE CONCRETE STRENGTHS TABLE FOR MINIMUM LD, WHICHEVER IS GREATER.
- COMPRESSION LAP SPLICE LENGTH SHALL BE 30db MINIMUM WHERE db IS THE DIAMETER OF BAR.
- WALL AND SLAB REINFORCEMENT SPLICES SHALL BE MADE SUCH THAT THE REQUIRED DISTANCES TO THE FACE OF WALL, SLAB OR REFERENCE LINES ARE MAINTAINED.
- SPLICES FOR ADJACENT REBARS SHALL BE STAGGERED MIN 600MM. AS PER ACI 318.
- ALL BARS MARKED CONTINUOUS SHALL BE PROPERLY LAPPED AT SPLICES AND CORNERS AND HOOKED AT NON CONTINUOUS ENDS.
- WELDING OF REINFORCING BARS WHERE PERMITTED SHALL BE PERFORMED IN ACCORDANCE WITH AWS D 1.4, STRUCTURAL WELDING CODE.
-
- ALL STANDARD HOOKS, CROSS TIES MINIMUM INSIDE BEND DIAMETERS AND DOWEL EMBEDMENT LENGTH SHALL BE AS PER ACI 318 SECTION 7.1.
- EXPLANATION OF NOTATION FOR REINFORCEMENT USED ON DRAWINGS:
EXAMPLE: D24-300B1.
D24 BAR GRADE 'D' IDEFORMED BAR) AND DIAMETER '24'
300 DISTANCE BETWEEN BAR CENTRES
B1 DESIGNATION OF REINFORCEMENT LAYER
WHENEVER SPECIFIED (REFER DIAGRAM 1)

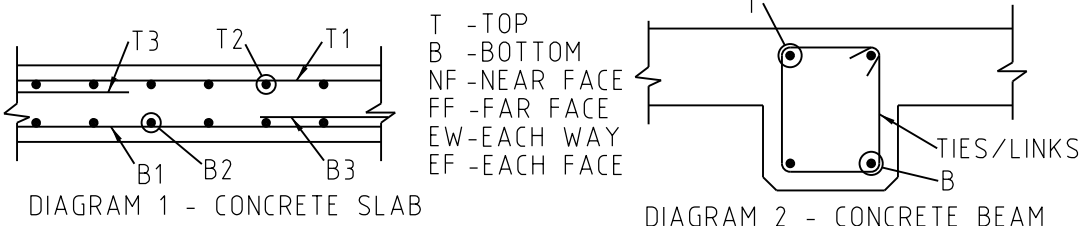


DIAGRAM 2 - CONCRETE BEAM



DIAGRAM 4 - CONCRETE COLUMN

- MINIMUM CONCRETE PROTECTIVE COVER FOR REINFORCEMENT SHALL BE IN ACCORDANCE WITH SBC 304 OR AS PER TABLE-2 WHICHEVER IS GREATER.
- DEVELOPMENT LENGTH SHALL BE REDUCED TO 0.8 LD AS CALCULATED FROM ABOVE TABLE-1 FOR BARS OF DIAMETER LESS THAN 20mm.

TABLE-2

No.	CONCRETE EXPOSURE CONDITION	CONCRETE COVER (mm)	REMARKS
1	EXPOSED TO SEA, WATER OR SOIL		
a	CONCRETE POURED AGAINST THE GROUND PERMANENTLY EXPOSED TO EARTH	NOT ALLOWED	
b	UNDERGROUND FOUNDATION, TIE-BEAMS, WALLS AND UNDERSIDE/SIDE OF SLABS	75	
C	TOP SIDE OF SLABS	50	WITH FORMED SURFACES
2	EXPOSED TO WEATHER		
a	UNDERGROUND AND ABOVE GROUND FOUNDATIONS	75	
b	COLUMNS & BEAMS	40	FOR STRUCTURE AND BUILDING ABOVE GROUND
c	SLAB,	25	
d	WALL & JOISTS	40	
e	ALL WELDED WIRE FABRIC	20	FOR 100MM THICK PAVEMENT
3	PRE-CAST CONCRETE ELEMENTS (MANUFACTURED UNDER PLANT CONDITIONS) CONCRETE EXPOSED TO EARTH OR WEATHER		
a	WALL PANELS PRESTRESSING TENDONS LARGER THAN 40MM DIA. PRESTRESSING TENDONS 40MM AND SMALLER IN DIA.	40 20	
b	OTHER MEMBERS PRESTRESSING TENDONS LARGER THAN 40MM PRESTRESSING TENDONS LARGER THAN 16MM TO 40MM DIA. PRESTRESSING TENDONS LARGER THAN 16MM DIA AND SMALLER	50 40 30	
	CONCRETE NOT EXPOSED TO EARTH OR WEATHER		
a	SLAB, WALL & JOISTS PRESTRESSING TENDONS LARGER THAN 40MM DIA. PRESTRESSING TENDONS 40MM AND SMALLER IN DIA.	30 20	
b	BEAMS, COLUMNS PRIMARY REINFORCEMENT TIES, STIRRUPS, SPIRALS	db, Where 16<db<40 10	

TABLE-3

BEND DIAMETER AND STANDARD HOOK LENGTHS:			
BAR DIA	BEND DIA	STANDARD HOOK	
		90° BEND 12db	180° BEND 12db
10	60	120	65
12	75	145	65
16	100	195	65
20	120	240	80
25	150	300	100
32	260	385	130

TABLE-4

BEND DIAMETER AND HOOK LENGTHS FOR STIRRUPS/TIES (mm).			
BAR DIA	BEND DIA	STIRRUP & TIE-HOOKS	
		90° BEND	135° BEND
10	60	60	60
12	75	75	75

- REINFORCEMENT SHALL BE SUPPORTED BY:
-PROPRIETARY CONCRETE SPACERS FOR FOOTINGS, GROUND SLABS AND FOUNDATIONS.
-STEEL/PLASTIC BAR CHAIRS FOR SUSPENDED SLABS, SIDE COVERS ON PEDESTAL FOOTINGS, BASES OR ANY VERTICAL SURFACES.
-BENT DEFORMED BAR SPACERS BETWEEN THE TOP AND BOTTOM REINFORCEMENT MATS.
- ANCHOR BOLTS
6.1 ANCHOR BOLT SPECIFICATION SHALL BE IN ACCORDANCE WITH MD-520-1000-EG-CV-SPC-0004 AND SHALL BE OF ROUND BAR MATERIAL MINIMUM GRADE 36(UND IN DESIGN DRAWING) STEEL TO ASTM F1554-STANDARD ANCHOR BOLT DETAILS REFER DRAWING MD-520-1000-EG-ST-STD-0005
6.2 ANCHOR BOLTS, NUTS AND WASHERS SHALL BE HOT DIPPED GALVANIZED IN ACCORDANCE WITH ASTM A153M
6.3 ALL ANCHOR BOLTS SHALL HAVE ONE WASHER AND MINIMUM ONE NUT. REFER TO ANCHOR BOLT STANDARD DRAWING MD-520-1000-EG-ST-STD-0005.
6.4 RESIN GROUTING OF ANCHOR BOLTS AND/OR DOWEL BARS INTO PERFORMED OR DRILLED HOLES SHALL BE CARRIED OUT USING AN EMPLOYER APPROVED INJECTION SYSTEM. INSTALLED STRICTLY IN ACCORDANCE WITH THE MANUFACTURER'S RECOMMENDATIONS.
6.5 ALL FABRICATED ANCHOR BOLTS SHALL BE TIGHTENED TO THE SNUG TIGHT CONDITION. ALL PROPRIETARY ANCHOR BOLTS SHALL BE TIGHTENED TO THE MANUFACTURER'S RECOMMENDED TORQUE.
6.6 ALL MAJOR ANCHOR BOLTS (E.G. FOR BUILDING COLUMNS, TRESTLE COLUMNS TANK COLUMNS, PIPE RACKS) SHALL BE CAST INTO CONCRETE. THE MINIMUM SIZE SHALL BE 20MM DIAMETER.
6.7 CHEMICAL ANCHORS (16MM MINIMUM) MAY BE USED FOR MINOR ITEMS. (STAIR STRINGERS, LADDER STRINGER, MINOR PIPE SUPPORTS). MECHANICAL ANCHORS SHALL NOT BE USED.
6.8 NON-SHRINK GROUT OF MINIMUM 50mm THICK SHALL BE PROVIDED UNDER ALL EQUIPMENT BASES WHERE SPECIFIC RECOMMENDATION NOT PROVIDED BY THE EQUIPMENT SUPPLIER, AND A MINIMUM 30mm THICK NON-SHRINK CEMENTITIOUS GROUT SHALL BE USED UNDER ALL OTHER STRUCTURAL BASE PLATES.

7.0 PENETRATIONS AND CAST IN ITEMS

- ALL EMBEDDED ITEMS SHALL BE IN ACCORDANCE WITH STANDARD DRAWING MD-520-1000-EG-CV-STD-0008 TO 0011.
- ALL EMBEDDED STEELWORK OTHER THAN REINFORCEMENT SHALL BE GRADE 250 OR 300 AND SHALL BE HOT DIPPED GALVANIZED.
- ALL CAST-IN PLACE STEELWORK SHALL BE FIRMLY SECURED TO FORMWORK SUCH THAT ALL EXPOSED STEEL SURFACES REMAIN IN PLACE DURING THE CONCRETE POUR. INSERTS EXPOSED SURFACE SHALL BE PAINTED AFTER ATTACHMENT OF ANY STRUCTURAL MEMBERS
- BEFORE POURING CONCRETE THE CONTRACTOR SHALL VERIFY REQUIREMENTS FOR ALL RELEVANT SERVICES, EMBEDDED ITEMS, BOLTS, SLEEVES, ETC.
- NO HOLES, CHASES OR EMBEDMENTS OTHER THAN THOSE SHOWN ON THE DRAWINGS SHALL BE MADE IN THE CONCRETE WITHOUT PRIOR APPROVAL OF THE EMPLOYER.
- ALL SLABS SHALL HAVE FULLY ANCHORED RE-ENTRANT BARS AT ALL RE-ENTRANT CORNERS AS SHOWN ON THE DESIGN DRAWINGS OR STANDARD DRAWING MD-520-1000-EG-CV-STD-0007.



DRAWN	M.UMAIR	09/14/23
CHECKED	P.KALYAN	09/14/23
ENGINEER	S.BISTA	09/14/23
APPROVED	A.SALAM	09/14/23
EMPLOYER	MAADEN	09/14/23
SCALE	NTS	A1

MA'ADEN PHOSPHATE-3 PHASE-1
RAS AL-KHAIR & WAAD AL SHAMAL
STRUCTURAL STANDARDS
CONCRETE GENERAL NOTES

DRAWING NUMBER MD-520-1000-EG-CV-STD-0001	REV C01
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